

One-Semester Real Analysis

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To the Student

This course guide will serve as the only reference to the concepts generally allowed for the course. As we encounter them, all relevant definitions and problems will appear, along with any guides to exploration. Pages should be printed out as they appear, but please feel free to make *lots* of notes!

A **Problem** is something to be proved. Some will be very easy, a few extremely hard. Most will be somewhere in between, to enhance your senses of confidence and accomplishment. Most class time will be hearing and responding to your solutions to these items; they are the core of the course.

A **Definition** is just that. However, in this type of course, it is extremely important to use definitions very accurately, and we try to use as few as possible to avoid confusion and oversaturation. Hence, they are highlighted just as much as problems.

An **Example** is not at the same level of importance as a problem, but it should usually be very helpful in understanding the content (and hopefully proofs) of the problems. It may be a worked-out example provided in class, or it may be some instructions for trying your own examples.

Chapter 1

Preliminaries and Review

In this course, we do assume certain background knowledge. This section is intended to review some of this, as well as to focus on specific items which may prove of use later.

1.1 Absolute Basics

There are some things I will not expand on at all.

We will take most normal mathematical and logical terminology from a proof transition course for granted.

We assume that sets exist. We will take the sets of counting ($\mathbb{N}$), integer ($\mathbb{Z}$), rational ($\mathbb{Q}$), and real ($\mathbb{R}$) numbers as known. In this course $\mathbb{N}$ starts with 1, not 0. We will denote the irrational numbers by $\mathbb{Q}^C$, a notation for the *complement* of the rationals with respect to the set $\mathbb{R}$.

In these sets we will assume the most basic properties of addition, subtraction, and multiplication, such as commutativity and distributivity. Likewise, we will later assume without proof basic facts about ordering numbers. At this time we explicitly assume that $0 < 1$ and that there are no integers between them. You will be able to use these facts to *prove* that is true for other consecutive pairs of integers. (The fact $0 < 1$ *does* require proof! Typically one uses a Well-Ordering type axiom.)

Although there is a lot of fun to be had with infinity, I do not foresee doing much with it other than using generic and specific infinite sets.

The rest of this preliminary chapter is intended to remind us of things for which I do think review could be useful.

1.2 Logic Review

In basic proof logic, we will usually be proving a statement of the nature

If P is true, then Q is also true.

Example 1. *Prove that if x is an even integer, then x^2 is also an even integer.*

However, there are more interesting examples of how to prove things. Recall the *converse* of the generic statement

If Q is true, then P is also true

and the *contrapositive* thereof

If Q is not true, then P is also not true.

Example 2. *State the converse of the previous example, and then its contrapositive. Prove that final statement.*

In (normal) mathematical logic, we say that the contrapositive and the original statement are *logically equivalent*. So you should use whichever one you find useful in proving theorems!

However, that isn't the case with the converse and the original statement.

Example 3. *Take the statement “If n is a prime number, then n is a positive integer” and give a reason why it is true, but its converse is false.*

In the first two examples, we have the highly unusual situation where both $P \implies Q$ and $Q \implies P$ are both true. We say that this is a *biconditional* statement and write $P \iff Q$, saying

P is true if and only if Q is true,

or “ P is necessary and sufficient for Q ”, or some variation thereon.

There are a few other tricky logic issues that we should definitely practice.

- *Or* is a word that means many things. In most of math (and here), “A or B” means “A or B or *both*”. This is important in statements like

If x or y is even, then xy is also even

since even if x and y were both even, xy would still be even.

Example 4. *Come up with a theorem which is not true with this meaning of ‘or’ but is true with the so-called exclusive or.*

- The other remaining tricky issue is that of quantifiers – things like “for all” and “there exists”. This is best tackled by trying it.

Example 5. *Restate these as accurately as possible in terms of logic, and then do all of the following with as few double negatives as possible, using natural negations if possible:*

- Negate the statement, “All professors at my college who live in my town are old and grey.”
- Negate the statement, “There exists a student at my college who does not live in my town,” in the most natural way possible (i.e., without double negatives).
- Give the contrapositive of the statement, “If there exists a professor at my college who is neither old nor grey, then all students at my college are young and bald.”

Problem 1. *Formally negate the statement, “For every positive number p , there is a positive integer N such that if an integer I is greater than N , then the reciprocal of I is less than p ,” with as few negatives and as naturally as possible. (But don’t try to prove it!)*

1.3 Sets and Functions

Recall that A is a *subset* of B (written $A \subset B$) if every element $x \in A$ is also an element of B . When a set is defined as a subset of another set with a given property, we often use “set builder” notation, such as

$$\{n \in \mathbb{Z} \mid 2 \text{ divides } n\}$$

for defining the even numbers. If a set has no elements, such as

$$\{P \text{ is a president of the USA before 2000} \mid P \text{ is female}\},$$

we call it the empty set ($\emptyset$). The familiar intersection ($\cap$) and union ($\cup$) operators are very useful. We distinguish between $\subset$ and $\subsetneq$, where in the latter the sets are not equal. Finally, we define $A \setminus B$ to be the set of all elements in A which are *not* in B .

Example 6. *Prove that $\mathbb{Z} \subset \mathbb{Q}$.*

Example 7. *Decide whether $\mathbb{N} \cup \{-n \mid n \in \mathbb{N}\} = \mathbb{Z}$.*

Example 8. *Find three nonempty sets A, B, C such that $A \cap B = \emptyset$, $B \subset C$, and $A \not\subset C$.*

Example 9. *Give an example where $A \neq B$ but $A \setminus B = \emptyset$.*

More interestingly, we can define infinite combinations of these things, if we have a collection of sets X_α , one for each element $\alpha \in A$ an infinite set.

Problem 2. Find (nonempty) sets $S_i \subset \mathbb{N}$ indexed by $i \in \mathbb{N}$ such that $S_{i+1} \subsetneq S_i$ and

$$\bigcap_{i=1}^{\infty} S_i = \emptyset.$$

Problem 3. Find (nonempty) sets $S_i \subset \mathbb{N}$ indexed by $i \in \mathbb{N}$ such that $S_i \subsetneq S_{i+1}$ but

$$\bigcup_{i=1}^{\infty} S_i \neq \mathbb{N}.$$

We should also recall that, for us, a function is a set.

Definition 10. A function from a set A to a set B is a subset of the Cartesian product $A \times B$ such that there is precisely one $(a, b) \in f$ for each $a \in A$. We call A the domain, and say that $f(a) = b$ if $(a, b) \in f$. There is no standard name for B , but if you insist on having one, use codomain.

Example 11. Write some function from $\mathbb{Z}$ to itself this way.

Recall a few properties functions might have.

- A function from A to B is *onto* if, for each $b \in B$, there is an $a \in A$ such that $f(a) = b$.
- A function from A to B is *one-to-one* if when $a \neq a' \in A$, then $f(a) \neq f(a') \in B$.
- A function from A to B is a *bijection* if it is one-to-one and onto.

Example 12. Suppose that $f(x) = x^2$, where $A = B = \mathbb{Z}$. Change A or B so that it is one-to-one. Find a way to change it to be onto.

Example 13. Find a bijection from $\mathbb{Z}$ to itself that is not the identity. If you remember what an inverse function is, find that, too.

There are two more important sets relevant to a given function.

Definition 14. Let $f : A \rightarrow B$ be a function.

- If $S \subset A$, we call the image of S under f , denoted $f(S)$, the set

$$\{f(x) \mid x \in S\}.$$

- If $T \subset B$, we call the preimage of T under f , denoted $f^{-1}(T)$, the set

$$\{x \in A \mid f(x) \in T\}.$$

Notice that the preimage exists whether or not the inverse function exists (and the notation f^{-1} refers only to the preimage in this course). We call the image of the entire domain the *image of the function*, $f(A)$.

Example 15. If $f(x) = x^2$ from $\mathbb{Z}$ to itself, find $f(\{-2, -1, 0, 1, 2\})$ and $f^{-1}(\{0, 1, 4\})$.

Problem 4. Find functions f and g and sets S and T such that $f(f^{-1}(T)) \neq T$ and $g^{-1}(g(S)) \neq S$. (Note that this is not composition!)

Problem 5. With all sets appropriately defined, only one of two following statements is necessarily true. Prove one and disprove the other.

$$\begin{aligned} f(X \cap Y) &= f(X) \cap f(Y) \\ f^{-1}(X \cap Y) &= f^{-1}(X) \cap f^{-1}(Y) \end{aligned}$$

1.4 Order Properties of the Real Numbers

We are almost ready to start some actual analysis, where the functions and sets will be defined on the real numbers. (As mentioned earlier, we will tacitly assume that numbers in this course are real numbers, unless we state otherwise.) In order to do this, we need some facts about real numbers, including the following **Order Axioms** previously promised.

Given $a, b, c \in \mathbb{R}$:

Order Axiom 1 If $a < b$ and $b < c$ then $a < c$.

Order Axiom 2 If $a \neq b$ then either $a < b$ or $b < a$ (but not both!).

Order Axiom 3 If $a < b$, then $a + c < b + c$.

Order Axiom 4 If $a < b$ and $c > 0$, then $ac < bc$; if $c < 0$, then $ac > bc$.

Order Axiom 5 Given $a \in \mathbb{R}$, then there is an integer n such that $n \leq a < n + 1$.

Example 16. Why are n and $n + 1$ are the smallest possible such integers?

Definition 17. Given these axioms, we say that the real numbers $\mathbb{R}$ are linearly ordered because “less than” and “greater than” have meaning.

The last axiom is sometimes called ‘Archimedean’. We call numbers greater than zero *positive*, and those greater than zero or equal to zero *nonnegative*; there is a similar definition for *negative* and *nonpositive*.

Adding the order axioms to the most basic facts about addition, subtraction, and multiplication already can allow us to prove slightly more advanced facts.

Example 18. Under these axioms, the various familiar relationships between positive, negative, addition, and multiplication are true – such as ‘a positive times a negative is negative’ and ‘a positive plus a positive is positive’.

Here is another useful fact related to the axioms. It may seem just as expected, but is quite different in style.

Problem 6. *Prove that for any positive $a \in \mathbb{R}$, there is an integer N such that $0 < \frac{1}{N} < a$.*

Finally, there is a special function which should be introduced now.

Definition 19. *Given $a \in \mathbb{R}$, we define the absolute value of a , denoted $|a|$, to be a if $a \geq 0$ and $-a$ otherwise.*

Example 20. *Show that $|-4|$ is what you expect, using this definition.*

Example 21. *Show that $|a| \geq 0$, with equality only if $a = 0$.*

Example 22. *Show that $|a|^2 = a^2$.*

The next two statements are pretty useful, and best proved (for now) by cases, though they have more clever proofs too.

Problem 7. *Show that $|ab| = |a||b|$.*

Problem 8. *Show that $|a + b| \leq |a| + |b|$.*

The latter is usually called the *triangle inequality*, and we have seen it before. A related statement is sometimes called the *reverse triangle inequality*.

Problem 9. *Show that $|a - b| \geq ||a| - |b||$.*

To prove the following, be sure to review properties of the integers we assume without proof.

Problem 10. *Assume that there is a positive element of the preimage of $\{2\}$ under the function $f(x) = x^2$ from the reals to the reals; that is, assume $\sqrt{2}$ exists. Show $1 < \sqrt{2} < 2$.*

We do assume here that in a proof transition or other experience you have seen that $\sqrt{2}$ is not a rational number, so now we have an irrational between two rationals, by a previous example.

The following problems are a good challenge to generalize this. You may need some other typical facts from an introductory proof course.

Problem 11. *Prove that between any two distinct real numbers there is a rational number.*

Problem 12. *Prove that between any two distinct real numbers there is an irrational number.*

We now turn to the main matters under consideration in this course.

Chapter 2

Point Sets and Sequences

Definition 23. A point is simply an element of the real numbers $\mathbb{R}$.

Definition 24. A point set is a nonempty subset of the real numbers.

You might well wonder why we need extra terms for these concepts. The reasons are mainly historical. Not only do these ideas undergird much more abstract ideas of sets, but in addition, there is a tradition worth preserving of “Moore method” courses using this terminology since we are focusing more on the *geometric* properties of the real line in this chapter.

One of the interesting things about choosing to look at the whole real number line is that we have two additional useful axioms. They seem nearly trivial, but are very important.

Point Set Axiom 1 If p is a point, then there is a point less than p and a point greater than p .

Point Set Axiom 2 If $p \neq q$ are two points, there is a point between them.

Note that these are closely related to the last few problems – but we are starting from scratch here, and you should treat these as the primary axioms we need for now.

Example 25. If instead of $\mathbb{R}$ we chose as our ‘background set’ some subset C of $\mathbb{R}$, these axioms might not hold. We can see this with $C = [0, 1] \cup [2, 3]$, with $p = 0$ in the first case and $p = 1, q = 2$ in the second.

Definition 26. We say that a point set O is a (finite) open interval if there are points $a \neq b$ such that O is the point set consisting of all points between a and b . That is,

$$O = \{x \mid a < x < b\} = (a, b).$$

Definition 27. We say that a point set I is a (finite) closed interval if there are points $a \neq b$ such that I is the point set consisting of a , b , and all points between a and b . That is,

$$I = \{x \mid a \leq x \leq b\} = [a, b].$$

In either case, a and b are called the *endpoints* of the interval.

Example 28. Give some examples of point sets that are not intervals.

Definition 29. If M is a point set, we say that p is an accumulation point of M if **every** open interval containing p also contains a point of M different from p .

To word this in set notation, p is an accumulation point of M if, for every open interval O with $p \in O$, there is $x \in O$ such that $x \in M \setminus \{p\}$.

Problem 13. Show that if M is an open interval and $p \in M$, then p is an accumulation point of M .

Problem 14. Show that if M is a closed interval and $p \notin M$, then p is not an accumulation point of M .

Problem 15. Determine whether the endpoints of an open interval M are accumulation points of M .

It is worth exploring exactly how many points it is possible or impossible for M to have. The next two problems are just a start in investigating that.

Problem 16. Show that if M is a point set having an accumulation point, then M contains (at least) two points. Determine whether M must contain at least 3 points.

Problem 17. Consider $\mathbb{Z}$ as a point set; show that it has no accumulation points.

Problem 18. Given that H and K are point sets, show either of the following:

- If p is an accumulation point of $H \cap K$, then p is an accumulation point of both H and K .
- If p is an accumulation point of $H \cup K$, then p is an accumulation point of H or p is an accumulation point of K .

Problem 19. If M is the set of all reciprocals of elements of $\mathbb{N}$, then zero is an accumulation point of M .

We will now start to turn to connecting the concepts of point sets to more familiar ones from calculus, beginning with sequences.

Definition 30. A sequence a is a function from $\mathbb{N}$ to $\mathbb{R}$.

Sometimes these are called *point sequences* to distinguish them from other potential sequences.

Even though technically the sequence is a function a , for any given $i \in \mathbb{N}$ we will write $a_i = a(i)$. Then we may write a sequence $a(1), a(2), a(3), \dots$

as $a_1, a_2, a_3, \dots$ – or, more formally, $\{a_i\}_{i=1}^\infty$. In order to keep the index visible (this is especially useful when we combine sequences), we will occasionally abuse notation slightly and write $\{a_i\}_{i=1}^\infty$ when referring to the function or the image set $\{a(i) \mid i \in \mathbb{N}\}$.

Example 31. Write down several sequences a you are familiar with. If possible, give an algebraic formula for each a_i in terms of i .

Example 32. Give an example where the image set of a sequence $\{a_i\}_{i=1}^\infty$ is finite.

There is a deep connection between sequences and accumulation points, which the next few problems will elucidate. First, a definition – one you may have seen in calculus in a different form.

Definition 33. We say that the (point) sequence $p = \{p_i\}_{i=1}^\infty$ converges to the point x if, given an open interval S containing x , there exists an $N \in \mathbb{N}$ such that if $n \geq N$ is also a positive integer then $p_n \in S$.

We simply say that p converges if there exists a point x to which the sequence converges. One could even write $p \rightarrow x$, informally. If a sequence does not converge to any point x , then we say it *diverges*.

The first problem about this should be used as a place to test ideas for how to prove convergence. Make sure you remember *all* the axioms you've learned and facts you've proved about real numbers – you may need them!

Problem 20. Consider the sequence given by $p_n = \frac{1}{n}$ (remember, $n \in \mathbb{N}$ is part of the definition of a sequence). Show that $p = \{p_i\}_{i=1}^\infty$ converges to 0.

Example 34. Before tackling any of the next few problems, try actually writing down the first 10 or 12 elements of each sequence.

Problem 21. Consider the sequence given by $p_n = 1 - \frac{1}{n}$. Show that p converges to 1.

Problem 22. Consider the sequence with even terms $p_{2n} = \frac{1}{2n-1}$ and odd terms $p_{2n-1} = \frac{1}{2n}$. Show that p converges to 0.

Problem 23. Consider the sequence with odd terms $p_{2n-1} = \frac{1}{2n-1}$ and even terms $p_{2n} = 1 + \frac{1}{2n}$. Determine whether p converges to 0.

The following problem connects our two concepts. The most profound property of the real numbers is part of this connection, as we'll soon see.

Problem 24. Show that if p converges to the point x and for each $i \in \mathbb{N}$, $p_i \neq p_{i+1}$, then x is an accumulation point of the image set of $\{p_i\}_{i=1}^\infty$.

Example 35. Why do we need the restriction that $p_i \neq p_{i+1}$? Is this an absolutely necessary restriction for x to be an acc. pt. of the image set?

Problem 25. Show that the sequence $\{\frac{1}{i}\}_{i=1}^{\infty}$ does not converge to a point other than zero.

Problem 26. Show that if p converges to the point x and y is a point different from x , then p does not converge to y .

Basic familiar facts about sequence convergence are recalled next. In these proofs, you may have to think a little more explicitly about what the intervals around x look like in order to combine sequences. Try doing some examples with explicit numbers first, to get a sense of how to approach them.

Problem 27. Show that if c is a number and $p = \{p_i\}_{i=1}^{\infty}$ converges to x , then cp (which means what you think it does) converges to cx .

Problem 28. Show that if $q = \{q_i\}_{i=1}^{\infty}$ converges to y and $p = \{p_i\}_{i=1}^{\infty}$ converges to x , then $\{q_i + p_i\}_{i=1}^{\infty}$ (which means what you think it does) converges to $y + x$.

Products and quotients of sequences behave like you think they will as well, and you can use these facts in the rest of the notes. We will include one special case soon.

Now we introduce a few more definitions that will lead us to the key axiom for the real numbers – *completeness*. We'll continue to see interplay between sequences and sets.

Definition 36. We say that a point set M is bounded if M is a subset of some closed interval.

Definition 37. We say that a point set M is bounded above if there is a point z such that if $x \in M$ then $x \leq z$; such a point is an upper bound.

Example 38. The property of a point set M being bounded below and the notion of a lower bound are defined similarly; try defining them.

Example 39. Show that a point set (in $\mathbb{R}$) being bounded is the same as it being bounded above and below.

Example 40. Find all upper bounds for $(0, 1)$, $[0, 1]$, and $(0, 1) \cap \mathbb{Q}^C$ (irrationals between 0 and 1).

In the next problem, remember that we 'abuse notation' by using $\{p_i\}_{i=1}^{\infty}$ to mean more than one mathematical object.

Problem 29. If the sequence p converges to the point x , then the image set $\{p_i\}_{i=1}^{\infty}$ is bounded.

You can use this concept to prove some of the more difficult sequence convergence properties.

Problem 30. Show that if q converges to 0 and p converges to x , then $\{q_i \cdot p_i\}_{i=1}^{\infty}$ (which means what you think it does) converges to 0.

Now we start to approach the heart of why calculus works.

Definition 41. We say p is a least upper bound (or supremum) of a point set M if p is an upper bound of M and $p \leq q$ for any other upper bound q of M .

Example 42. Define the greatest lower bound/infimum by analogy.

Problem 31. Find the suprema of $(0, 1)$, and $(0, 1) \cap \mathbb{Q}^C$. If we could apply the definition of supremum to $\emptyset$, what would its supremum be?

Problem 32. Prove that the supremum of a point set is unique, if it exists.

Problem 33. If M and N are point sets with suprema, characterize the supremum of $M \cup N$.

If M and N are point sets, define cM to be $\{cx \mid x \in M\}$ and $M + N = \{x + y \mid x \in M, y \in N\}$.

Problem 34. Assuming M and N have suprema, prove either that $\sup(cM)$ is $c \sup(M)$ (given $c > 0$) or that $\sup(M + N) = \sup(M) + \sup(N)$.

Example 43. Show that $c \inf(M) = \sup(cM)$ if $c < 0$. What other properties are there relating $\inf$, $\sup$, and c ?

The reason the supremum is so important is this fundamental axiom:

Axiom – Completeness of the Real Numbers If M is a point set and is bounded above, then M has a supremum.

Example 44. Come up with a sequence p such that the image set is unbounded $\{p_i\}_{i=1}^{\infty}$ and hence does not have a supremum.

Example 45. Show that the axiom is not true if one requires that the supremum be a rational number.

It will be quite useful in the future to show that there is an equivalent way to formulate completeness in terms of sequences.

Definition 46. We say that a sequence p is nondecreasing if $p_i \leq p_{i+1}$ for all $i \in \mathbb{N}$. (The concept of nonincreasing is defined similarly.)

Example 47. Replace the $\leq$ above with $<$ to define the notion of (strictly) increasing. Find examples of nondecreasing sequences which are not increasing (and similarly for nonincreasing/decreasing).

Problem 35. If p is a nondecreasing sequence such that the image set $\{p_i\}_{i=1}^{\infty}$ is bounded above, then p converges to some point x .

Problem 36. Assuming the previous problem is true, prove the completeness axiom for point sets.

Why is all this so important? One reason is that we can use the completeness of the reals to *prove* one of the axioms we snuck in earlier. It may be thought of as the ‘real’ reason why the following is true, since open intervals can be as small as we need them to be.

Problem 37. *Using the previous problem (but not its proof!), show that for any point x , there is an $n \in \mathbb{Z}$ such that $n > x$. (Hint: contradiction.)*

A use of the *alternate* definition of completeness is proving properties of bounded sequences. The most famous of these is

The Bolzano-Weierstrass Theorem: Let us call a sequence $\{b_k\}_{k=1}^{\infty}$ a *subsequence* of another sequence $\{a_n\}_{n=1}^{\infty}$ if there is a sequence of natural numbers $\{n_i\}_{i=1}^{\infty}$ with $n_i < n_{i+1}$ such that $b_k = a_{n_k}$. Then every sequence with bounded image set has a convergent subsequence.

Chapter 3

Continuity

Definition 48. We say that a function f is continuous at a point x in its domain (or at the point $(x, f(x))$) if, for any open interval S containing $f(x)$, there is an open interval T containing x such that if $t \in T$ is in the domain of f , then $f(t) \in S$.

Definition 49. A function f is continuous if it is continuous at every point in its domain.

Let's show that this definition of continuity behaves the way we expect from calculus. The first example will make it particular easy to find T , given S .

Example 50. Show that $f(x) = x$ is continuous for all $x \in \mathbb{R}$.

Example 51. Pick one: Show that either $f(x) = 2x$ or $f(x) = x + 3$ is continuous for all $x \in \mathbb{R}$.

Problem 38. Show that any linear function is continuous for all $x \in \mathbb{R}$.

Problem 39. Let $f(x) = 1$ if $x \in [0, 1]$, and $f(x) = 0$ otherwise. Determine and prove exactly for which x this function is continuous.

Problem 40. Let $f(x) = x^2$. Show that f is continuous at $x = 2$. (You may assume that square roots exist.)

Problem 41. Define a function $g : \{0\} \rightarrow \mathbb{R}$ by $g(0) = 0$. Show that g is continuous at $x = 0$.

Problem 42. Let $f(x) = 1$ if $x \in \mathbb{Q}$, and $f(x) = 0$ otherwise. Find all points x where f is continuous.

It turns out that we can connect continuity to sequences, too.

Definition 52. We say that a function f is sequentially continuous at a point x if, for every sequence $\{x_i\}_{i=1}^{\infty}$ (in the domain of f) converging to x , it is also true that $\{f(x_i)\}_{i=1}^{\infty}$ converges to $f(x)$.

Problem 43. *If a function f is continuous at x , it is also sequentially continuous at x .*

Problem 44. *If f is sequentially continuous at x , it is continuous at x .*

So these concepts are *equivalent on the real numbers*. However, we still distinguish between them, since this isn't the case for all sets. See, for instance, <http://www.youtube.com/watch?v=sZ5fBHGyurg>.

The sequential way of thinking of continuity makes proving some basic facts easier. Be clear on the logic in the next two, and don't forget what we've already proved or stated about sequence convergence.

Problem 45. *If $c \in \mathbb{R}$ is a constant and if f and g are continuous at a point x , then so are the functions cf and $f + g$.*

Problem 46. *If f and g are continuous at a point x , so is the function fg .*

Example 53. *All polynomials in x are continuous for all x (here, in the full domain $\mathbb{R}$).*

Before finally moving to calculus proper, we'll deal briefly with two foundational theorems about $\mathbb{R}$ and continuous functions that have everything to do with completeness.

The Extreme Value Theorem: Let I be a closed interval and suppose that f is a function which is continuous on I . Then there exists $x_M \in I$ such that $f(x_M) \geq f(x)$ for all $x \in I$.

The Intermediate Value Theorem: Suppose f is a continuous function on a closed interval $I = [a, b]$, and that $f(a) < 0 < f(b)$. Then there is a point $x \in [a, b]$ such that $f(x) = 0$.

Problem 47. *Suppose f is a continuous function on a closed interval $I = [a, b]$ and that $f(a) < c < f(b)$. Then there is a point $x \in [a, b]$ such that $f(x) = c$.*

Note that this is also true if $f(b) < c < f(a)$. One way to get a handle on these facts is to find counterexamples when the hypotheses are not satisfied.

Problem 48. *Find a continuous function on $\mathbb{R}$ which does not satisfy the conclusions of the Extreme Value Theorem; do the same for a discontinuous function on $[0, 1]$.*

Problem 49. *Find a continuous function on $[0, 1] \cup [2, 3]$ with $f(0) = -1$, $f(3) = 1$ which does not satisfy the conclusions of the Intermediate Value Theorem; do the same for a discontinuous function on $[0, 1]$ where $f(0) = -1$, $f(1) = 1$.*

Proving the EVT is most often done via the Bolzano-Weierstrass Theorem and the sequential definition of continuity (or other definitions); it's rather tricky to do it using only the supremum definition. Here is one piece of the proof (this piece actually applies to non-continuous functions).

Problem 50. Let f be a function on $[0, 1]$, and assume that the image $f([0, 1])$ has a supremum. Show there is a sequence of points $\{x_i\}_{i=1}^{\infty}$ in $[0, 1]$ such that $\{f(x_i)\}_{i=1}^{\infty}$ converges to that supremum.

Example 54. Explain why that's not enough, by itself, to prove the whole Extreme Value Theorem.

Proving the IVT, on the other hand, can definitely be done using just the set definitions of completeness and continuity. The next problem should get your creative juices flowing as to how to find the x such that the IVT holds for that x . Be careful to show that your x exists!

Problem 51. Let f be a function on $[0, 1]$ such that $f(0) = -1$, $f(1) = 1$, and $f([0, 1]) = \{-1, 1\}$. Show (without using the IVT) that there is an $x \in [0, 1]$ such that f is not continuous at x .

Problem 52. Prove the Intermediate Value Theorem.

These two theorems combine in a deep way. If we had time for just a bit more topology, we could have introduced the words *compact* and *connected*; these more advanced concepts are necessary to describe the proper generalization of the following result for more general spaces than $\mathbb{R}$.

Problem 53. The image of a closed interval under a continuous function is either a closed interval or a single point.

Problem 54. Find a function f on a closed interval I such that $f(I)$ is also a closed interval, but for which f is not continuous for some $x \in I$. Find a continuous function g on a closed interval $I = [a, b]$ such that $g(a) = g(b)$ but $g(I)$ is not (just) the single point $g(a)$. Finally, find a function h on a closed interval I such that $h(I)$ is not a closed interval.

Chapter 4

Differentiation

It's time for calculus.

Definition 55. Let f be a function, and x be a point in the domain of f . Given a real number D , we say that f has derivative D at the point x if the following two conditions hold:

- The point x is an accumulation point of the domain of f .
- If S is an open interval containing D , then there is an open interval T containing x such that if $t \in T$, $t \neq x$, and t is in the domain of f , then

$$\frac{f(t) - f(x)}{t - x} \in S.$$

If f does indeed have a derivative at some points in its domain, we call the derivative of f , denoted f' , the function giving by sending such an x to the derivative D at the point x . We write $f'(x) = D$ in that case. We also say that f is differentiable at such points.

Note that the definition of derivative automatically excludes the kind of behavior we saw with continuous functions, where a function defined only at a single point was continuous.

Example 56. A function defined only on $\mathbb{Z}$ cannot have a derivative.

Example 57. Find and prove a formula for the derivative of $f(x) = 3$.

Problem 55. Find and prove a formula for the derivative of $g(x) = 2x - 5$.

Problem 56. Find the derivative of $h(x) = x^2 - x + 1$ at $x = 2$.

Problem 57. Find the derivative of $h(x) = x^2 + ax + b$ for any $a, b \in \mathbb{R}$.

Problem 58. If f is differentiable at x and $c \in \mathbb{R}$, show that the function cf also has a derivative at x and $(cf)'(x) = cf'(x)$.

Problem 59. If f and g are differentiable at x , show that the function $f + g$ also has a derivative at x and $(f + g)'(x) = f'(x) + g'(x)$.

When it comes to products, things are more subtle, as shown in stating the familiar product rule. On a different note, our facts about products from previous chapters can be used to show that differentiable functions are continuous. The following examples show the converse of this fact isn't true, and that even when f' does make sense, the domain of f' might be a proper subset of the domain of f .

Problem 60. Let $f(x) = |x|$. Prove that f is differentiable everywhere except at $x = 0$.

Problem 61. Let $f(x) = x$ if $x \in \mathbb{Q}$, and $f(x) = 0$ otherwise. Show that f is continuous at $x = 0$, but not differentiable.

Example 58. There is a function like this which is differentiable at $x = 0$!

It's useful to spend just a little time thinking about how one would prove the usual facts about derivatives used in a calculus course.

Problem 62. Suppose a function f defined on the closed interval $[a, b]$ has a derivative at some $x \in (a, b)$ and $f(x) \geq f(t)$ for all $t \in (a, b)$. Show that then $f'(x) = 0$.

So if there is an extreme value of a differentiable function, it occurs at a critical point. This leads us to (a variant of) Rolle's Theorem.

Problem 63. Suppose $a < b$, that f is differentiable on (a, b) , continuous on $[a, b]$, and that $f(a) = f(b) = 0$. Show there exists a point $c \in (a, b)$ such that $f'(c) = 0$.

Now consider a more general function where $f(a) \neq f(b)$. If one carefully considers the secant line connecting the endpoints of the graph of this function and applies the previous problem to this, one can prove the following crucial result, the **Mean Value Theorem**:

Problem 64. Suppose f is a continuous function on $[a, b]$ and differentiable on (a, b) . Then there is a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Among many other wonderful things this theorem provides, we have the following standard result.

Problem 65. If the derivative of a function is uniformly zero on a closed interval, then the function is constant on that interval.

Chapter 5

Integration

Unlike with differentiation, we will need a number of auxiliary definitions for beginning integration.

Definition 59. A set of points $\{t_0, t_1, \dots, t_n\}$ such that $a = t_0$, $b = t_n$, and $t_{i-1} < t_i$ for all i is called a partition of the closed interval $[a, b]$.

Example 60. Give some partitions of $[2, 4]$ and $[-1, 0]$. What would the points of a partition of $[0, 1]$ be if $t_i - t_{i-1}$ is the same for all i ?

Definition 61. Given $[a, b]$ and calling $\frac{b-a}{n} = \Delta t$, we may for convenience give the partition $\{a, a + \Delta t, a + 2\Delta t, \dots, b - \Delta t, b\}$ the name regular partition P_n (where $[a, b]$ is implicit).

Definition 62. Given a partition $P = \{t_0, t_1, \dots, t_n\}$ of $[a, b]$, let y_i be the supremum of $\{f(x) \mid x \in [t_{i-1}, t_i]\}$. We say that the sum

$$U_P(f) = \sum_{i=1}^n y_i(t_i - t_{i-1})$$

is the upper sum for P of f on $[a, b]$.

Definition 63. Given a partition $P = \{t_0, t_1, \dots, t_n\}$ of $[a, b]$, let z_i be the infimum of $\{f(x) \mid x \in [t_{i-1}, t_i]\}$. We say the sum

$$L_P(f) = \sum_{i=1}^n z_i(t_i - t_{i-1})$$

is the lower sum for P of f on $[a, b]$.

Definition 64. A bounded function is a function with a bounded image set.

We will assume all functions in this section are bounded.

Example 65. Give some upper and lower sums on some of these partitions with non-constant bounded functions.

Problem 66. Let $f(x) = 0$ for each number x in $[0, 1]$ except $x = 0$, and let $f(0) = 1$. Show that $U_P(f) > 0$ for all partitions of $[0, 1]$, but that for any positive number ε there is a partition P_ε such that $U_{P_\varepsilon}(f) < \varepsilon$. In addition, fully describe all lower sums of f on $[0, 1]$.

Problem 67. Let $f(x) = x$, and let P_n be the (regular) partition of $[0, 1]$ given by $\{0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}, 1\}$. Compute $U_{P_n}(f)$ and $L_{P_n}(f)$, and then give a formula for $U_{P_n}(f)$ and $L_{P_n}(f)$.

Problem 68. Suppose that f is bounded, with m a lower bound and M an upper bound of $f([a, b])$. Show that, for any partition P of $[a, b]$, $U_P(f) \leq M(b - a)$ and $L_P(f) \geq m(b - a)$.

Problem 69. Suppose that f is bounded on $[a, b]$ and P is a partition of $[a, b]$. Show that $L_P(f) \leq U_P(f)$.

Note that as a result of the previous problems that the set of all upper and lower sums of f is a bounded point set.

Definition 66. We say that a partition Q refines (or is a refinement of) a partition P if P and Q are both partitions of $[a, b]$ and $P \subset Q$.

Example 67. Take some partitions from previous examples, and refine them.

Problem 70. If P and P' are both partitions of $[a, b]$, there is a partition Q which is a refinement of both.

Problem 71. If f is a bounded function on $[a, b]$ and P and Q are partitions of $[a, b]$ such that Q refines P , show that $L_P(f) \leq L_Q(f)$ and $U_P(f) \geq U_Q(f)$.

We are nearly ready to define (definite) integrals.

Definition 68. With f a bounded function on $[a, b]$, we define the upper integral from a to b of f to be the infimum of the set of all upper sums for f on $[a, b]$. We denote the upper integral by $\int_a^b f$.

Definition 69. With f a bounded function on $[a, b]$, we define the lower integral from a to b of f to be the supremum of the set of all lower sums for f on $[a, b]$. We denote the lower integral by $\int_a^b f$.

Example 70. Why do the upper integral and lower integral always exist in this context?

Problem 72. With f as throughout this section, show that $\overline{\int_a^b f} \geq \underline{\int_a^b f}$.

Problem 73. If we let $f(x)$ be the function such that $f(x) = 1$ if $x \in \mathbb{Q}$ but $f(x) = 0$ otherwise, show that in this case $\overline{\int_0^1 f} > \underline{\int_0^1 f}$.

We can get more interesting conclusions if f has more properties.

Problem 74. Suppose that f is not just bounded, but continuous. Suppose further that $f(x) \geq 0$ for all $x \in [a, b]$ and that for some $z \in [a, b]$, $f(z) > 0$. Then show that $\int_a^b f > 0$.

Definition 71. For f as above, if $\overline{\int_a^b f} = \underline{\int_a^b f}$, then we say that f is integrable on $[a, b]$ and call this common value the integral of f from a to b . We denote the integral, if it exists, by

$$\int_a^b f \text{ or } \int_a^b f(x) dx.$$

Traditionally when this value exists we say that f is *Riemann* integrable and the number $\int_a^b f$ is the Riemann integral. Technically, however, it is the *Darboux* integral, with Riemann integrals coming from so-called Riemann sums. The two notions can be proved equivalent.

Since we saw above that the lower and upper integrals can be distinct, there exist nonintegrable bounded functions (in this sense). Luckily, there also exist integrable functions.

Example 72. Show that constant functions $f(x) = c$ are indeed integrable over any closed interval.

It is worth pondering why the information we have from previous examples is *not* quite enough to show that we have more integrable functions. After all, readers may be familiar with computing integrals. Historically, though, finding the values of many integrals (originally called *quadrature* since one wished to construct an actual square with the same area) was quite challenging. Since many applications in a variety of fields

A first step in this direction is the following characterization, which is very useful in showing functions are integrable.

Problem 75. Let f be a bounded function on $[a, b]$. Show that if f is integrable then for each positive number ε there is a partition P of $[a, b]$ such that $U_P(f) - L_P(f) < \varepsilon$, or show the converse (both are true and useful).

Problem 76. Pick one of these functions, show it is integrable on $[0, 1]$, and compute the integral. (Both functions are integrable; you may want to use previous problems.)

- The function f such that $f(x) = 0$ for each number x in $[0, 1]$ except $x = 0$, where $f(0) = 1$;
- The function $f(x) = x$.

This is not a lot of material to integrate. The next sets of theorems will vastly expand our repertoire of functions known to be integrable, though not yet of those with precise values at hand.

Definition 73. A function is nondecreasing if for each $x < y$ in the domain of f we have $f(x) \leq f(y)$. (We say that it is (strictly) increasing if $f(x) < f(y)$ always. The concept of nonincreasing is defined with $\geq$ in place of $\leq$.)

Example 74. Give several nondecreasing functions we haven't already used. (This is very similar to the equivalent sequence concept!)

Problem 77. Prove that a nondecreasing bounded function f on a closed interval $[a, b]$ is integrable.

You may want to write down a number of upper and lower sums to prepare for the previous problem. What previous problem will likely be easiest to use to prove integration theorems?

Problem 78. Prove that a nonincreasing bounded function f on a closed interval $[a, b]$ is integrable.

Example 75. Since $\sqrt{1-x^2}$ is nonincreasing on $[0, 1]$, this shows that $4 \int_0^1 \sqrt{1-x^2} dx$ exists. What number is this?

This example is clever, but still holds a bit of magic, since even a small change from $\sqrt{1-x^2}$ might make it very hard. Could we integrate in general?

The preceding results mean that power functions, exponentials, and many other standard functions are integrable on certain intervals, even if we can't finish computing them yet. The next theorems verify that most related functions are integrable on most typical intervals, and will help evaluate them too.

Problem 79. Let $c \in \mathbb{R}$ and let f be integrable on $[a, b]$. Show that cf is integrable on $[a, b]$ and $\int_a^b cf = c \int_a^b f$.

Problem 80. Let f and g be integrable on $[a, b]$. Show that $f+g$ is integrable on $[a, b]$. (Can you show the formula $\int_a^b (f+g) = \int_a^b f + \int_a^b g$?)

Not every possible statement is true, of course.

Problem 81. Give two functions f and g which are integrable on $[0, 1]$ such that fg is also integrable on $[0, 1]$ but $\left(\int_0^1 f\right)\left(\int_0^1 g\right) \neq \int_0^1 fg$.

We will now head straight for the punchline to the story of calculus, the 'fundamental theorems'. Just as these were not recognized as such until quite some time after Newton and Leibniz, you may want to re-evaluate what you think their importance is here.

We begin by proving some facts about the upper integral (they are also true for the lower integral). Remember, all functions are bounded.

Problem 82. Assume that $[a, b]$ is a closed interval and $c \in (a, b)$. Show that $\overline{\int_a^c f} + \overline{\int_c^b f} = \overline{\int_a^b f}$.

On a related note, for upper (and indeed for lower and Riemann) integrals, we define $\overline{\int_b^a f} = -\overline{\int_a^b f}$ and $\overline{\int_a^a f} = 0$. Can you see why we do this?

Problem 83. Show that if $f \leq g$ on the interval $[a, b]$, then $\overline{\int_a^b f} \leq \overline{\int_a^b g}$.

The next property is of course also true for the regular integral (when it exists), and is called the integral mean value theorem. Can you see why?

Problem 84. Let f be a continuous function on $[a, b]$; then there is a number $c \in [a, b]$ such that $\overline{\int_a^b f} = f(c)(b - a)$.

The next thing we do is *really* expand our collection of integrable functions!

Problem 85. Let U be defined such that for each $x \in [a, b]$, $U(x) = \overline{\int_a^x f}$, and let L be defined likewise but with $\underline{\int_a^x f}$. Show that if f is continuous on $[a, b]$, then U and L are differentiable, with $U'(x) = f(x) = L'(x)$.

Problem 86. In the previous problem, show that in fact $U(x) = L(x)$, and conclude that if f is continuous on $[a, b]$ then f is integrable on $[a, b]$.

And finally we get the most beloved theorems of all. Their true importance lies in the inverse relationship between derivatives and integrals, and that one can evaluate integrals (which a priori come from the sums we did earlier) exactly using information about derivatives. You should be able to prove these without lots of subtle work, using the facts about lower and upper integrals.

Problem 87. Let f be a continuous function on $[a, b]$, and let F be defined such that for each $x \in [a, b]$, $F(x) = \int_a^x f$. Then for each point $x \in [a, b]$, F has a derivative at x and we can compute $F'(x) = f(x)$.

Problem 88. Suppose that g is a function on $[a, b]$, that g has a derivative at each point of $[a, b]$, and that the function g' is itself continuous at each point in $[a, b]$. Then show that $\int_a^b g' = g(b) - g(a)$.

These are the keys which made the efforts of Newton and Leibniz into the *calculus* – something one could calculate with. It is worth noting that this final theorem is true even if f' is just integrable – truly a great theorem.